

Toward a Tractability Frontier for Exact Relevance Certification

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Abstract

Exact relevance certification asks which coordinates are necessary to determine the optimal action in a coordinate-structured decision problem. We first give a positive structural consolidation: the tractable families studied here split into core mechanisms, regime lifts, and degenerate collapses, yielding a finite primitive basis. We then prove a maximal realizability barrier: arbitrary labeling kernels are optimizer-realizable, so quotient shape alone cannot characterize a tractability frontier.

The main new result is a meta-impossibility theorem for a principled admissibility class. Admissible predicates are required to be polynomial-time checkable, structural, and invariant under closure laws already forced by exact-certification semantics (relabelings, positive affine transformations, and duplication/irrelevance operations). Closure-law invariance in this class is grounded not only in those internal closure laws, but also in structural convergence: zero-distortion summaries, quotient entropy bounds, and Fisher-style support counts already factor through the same optimizer-quotient core. Within this class, we build explicit same-orbit disagreement witnesses for four obstruction families—dominant-pair concentration, margin masking, ghost-action concentration, and additive/statewise offset concentration. Each witness uses an action-independent, pair-targeted affine term to move within a closure orbit while changing obstruction status. Consequently, no admissible normalization predicate can provide an exact tractability characterization over these families. This yields an admissible-collapse theorem and places the frontier program in a Rice/Arrow-style meta-impossibility tradition: once structural invariances are accepted, the sought classifier does not exist in that form.

1 Introduction

Exact relevance certification asks: given a coordinate-structured decision problem, which coordinates are necessary to determine the optimal action set? This question appears whenever one seeks principled feature deletion or abstraction while preserving optimization behavior. Under succinct encodings, exact certification is hard in the ambient regimes (coNP-complete in static settings and PSPACE-complete in sequential settings [29]), so any tractability statement must be genuinely structural.

This paper complements the optimizer-quotient and certification-trilemma results of [29]. That work isolates the semantic object on which exact certification depends and maps the surrounding complexity landscape across regimes. The present paper asks whether that semantic core admits a finite structural tractability classifier at the representation level. The answer is negative for the admissibility class defined below.

The current landscape is already rich enough to separate two distinct tasks. The first is “positive consolidation”: identify and organize known polynomial-time islands. The second is “meta-frontier

control”: determine whether a finite structural classifier can exist at all once one respects the invariances forced by exact-certification semantics. This paper addresses both tasks.

On the positive side, we prove that the tractable families treated here split into three roles: primitive core mechanisms, regime lifts, and degenerate collapses. This yields a finite primitive basis. On its own, however, this does not establish a frontier theorem: finite inventory is not finite characterization.

The natural next move is to classify by optimizer-induced quotient structure. We show this is impossible in the unconstrained sense: optimizer realizability is maximal, and arbitrary labeling kernels are realizable. Thus quotient shape is too expressive to carry the boundary by itself.

This exactness requirement is not arbitrary. At zero distortion, any summary that preserves optimal actions must still separate distinct optimizer classes: if two different optimizer classes were assigned the same summary value, the summary would merge states with different optimal-action sets and would cease to be lossless. Exact relevance therefore marks the theoretical floor for lossless abstraction. One may study approximate summaries for other purposes, but once the goal is exact decision preservation, the optimizer quotient is already the unavoidable reference object.

This realizability barrier motivates the organizing principle of the paper: tractability predicates must be evaluated on representation-level structural classes, but modulo theorem-forced equivalences of presentation. In particular, closure under relabelings, positive affine utility reparameterizations, duplication, and irrelevant-coordinate extension is not a modeling preference; it is inherited from the fact that exact certification depends only on the induced decision quotient relation.

Standing Semantics

For a decision problem with state space S , action space A , and utility U , write

$$\text{Opt}(s) = \{a \in A : U(a, s) \text{ is maximal among all actions at } s\}.$$

This induces the decision-equivalence relation

$$s \sim_{\text{Opt}} s' \iff \text{Opt}(s) = \text{Opt}(s').$$

The associated optimizer quotient is the quotient set $S/\sim_{\text{Opt}}$. A coordinate set I is *sufficient* when agreement on I forces equality of optimizer classes, and a coordinate is *relevant* when deleting it destroys sufficiency. These are the semantic objects studied throughout the paper; later closure laws and obstruction families are all phrased relative to this quotient viewpoint.

At the same time, unrestricted predicates trivialize any impossibility theorem via oracle encoding. We therefore isolate a principled admissibility layer: polynomial-time checkability, structural extractability, closure-law invariance, and bounded-pattern definability. This class excludes direct semantic encoding while still covering the structural results established here.

Within this admissibility class, we prove a four-family no-go template. For each obstruction family we construct two slices with different obstruction status in the same closure orbit. The key mechanism is an action-independent, pair-targeted additive affine term (the statewise “ α -component” of positive affine transport), which changes the obstruction predicate while preserving closure equivalence. Closure-law invariance then transports any candidate predicate across the orbit and forces contradiction.

The dominant-pair family is the canonical instance, and the same mechanism now applies to margin masking, ghost-action concentration, and additive/statewise offset concentration. The resulting theorem states that no admissible normalization predicate can serve as a tractability characterization in the required sense.

Conceptually, the result belongs to the same meta-impossibility lineage as Rice-style and Arrow-style theorems: once one fixes a natural invariance package and a nontrivial expressivity domain, the desired classifier does not exist. The open problem is to identify a stronger structural principle, if one exists, that yields a correct frontier theorem.

Roadmap. Section 2 gives the finite-basis consolidation for the current positive landscape; Section 3 separates lifts from degeneracies; Section 4 formulates the frontier question and its obstructions; Section 5 proves the realizability barrier; Section 6 develops the necessary conditions and the admissible-collapse theorem; Section 7 positions the result in related meta-impossibility and tractability traditions; Section 8 concludes.

An accompanying Lean development mechanically verifies the realizability constructions, the orbit-gap template, and the family-level admissible no-go theorems used in the collapse argument. The mathematical exposition below is intended to be fully self-contained and does not rely on consulting the mechanization.

2 Finite Basis for the Current Positive Landscape

We distinguish between *tractable families treated here* and *primitive tractability mechanisms*. A tractable family is any named positive condition included in the present analysis. A primitive mechanism is a smaller conceptual source of tractability that may explain several families at once.

For the present analysis, the primitive list is explicit:

1. the six core structural subcases,
2. constant-optimizer collapse, and
3. finite explicit enumeration.

Thus the current primitive basis has size eight.

Theorem 2.1¹ (Explicit Partition of the Current Positive Landscape). *The fifteen tractable families considered here admit an explicit disjoint partition into three blocks:*

1. *six core structural families,*
2. *four lifted families that reduce to existing core subcases, and*
3. *five degenerate families that become tractable by optimizer collapse or finite explicit enumeration.*

Moreover, each family is classified into exactly one of these three blocks by an explicit classification map.

Proposition 2.2² (Explicit Inventory). *The fifteen tractable families considered here are exactly the following:*

1. *core: bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry;*
2. *lifts: product distribution, bounded support, bounded horizon, full observability;*
3. *degenerate: single action, bounded state space, strict global dominance, constant optimal set, multiplicative-separable constant-sign.*

¹ Lean: FR1-2 ² : FR25-28

Corollary 2.3³ (Finite Basis for the Current Positive Landscape). *Projecting the partition of Theorem 2.1 to primitive mechanisms yields an explicit finite list of eight primitive tractability mechanisms. The list consists of the six core structural subcases*

bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry, together with two degenerate mechanisms:

constant-optimizer collapse and finite explicit enumeration.

Proposition 2.4⁴ (Explicit Primitive Basis). *The current primitive basis consists exactly of bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry, constant-optimizer collapse, and finite explicit enumeration.*

Proposition 2.5⁵ (Partition Counts). *The three blocks from Theorem 2.1 have sizes*

6 (core), 4 (lifted), 5 (degenerate).

The current positive landscape therefore contains fifteen tractable families accounted for by eight primitive mechanisms.

Proposition 2.6⁶ (All Independent Core Mechanisms Are Nondegenerate). *The analysis already contains witness families showing that five core mechanisms are individually indispensable in the sense of basis coverage:*

- 1. a low-rank-only witness family,*
- 2. a separable-only witness family,*
- 3. a bounded-actions-only witness family,*
- 4. a bounded-treewidth-only witness family, and*
- 5. a symmetry-only witness family.*

Each of these families remains tractable by its designated mechanism while failing the corresponding versions of the other independent core mechanisms.

Proposition 2.6 settles the independent part of the basis question on the present surface. Every core mechanism that is not merely contained in another is shown to be genuinely needed.

More concretely, the present theory contains five distinct kinds of “only-by” witness:

- 1. low-rank only,*
- 2. separable only,*
- 3. bounded-actions only,*
- 4. bounded-treewidth only, and*
- 5. symmetry only.*

Proposition 2.7⁷ (Weak Tree Ordering Does Not Imply Width One). *There exists a dependency presentation satisfying the older monotone tree-order predicate **TreeStructured** whose dependency graph has real treewidth greater than one. Width-one containment requires the stricter parent-tree condition from Proposition 2.9, not merely monotone ordering.*

³ : FR3 ⁴ : FR29 ⁵ : FR2 ⁶ : FR101, FR105–106, FR108, FR112–113 ⁷ : FR114

Proposition 2.8⁸ (Cyclic Dependencies Recover Hardness). *The reduction family for exact relevance certification yields **coNP**-hard instances with cyclic dependency structure. Thus the tree-structured regime is not only sufficient for tractability; leaving it also reaches the hard side.*

Proposition 2.9⁹ (Parent-Tree Structure Gives Width-One Decompositions). *Suppose a dependency presentation is tree-structured in the strict parent sense: every dependency points to a smaller coordinate and every coordinate has at most one parent. Then the induced dependency graph admits a genuine tree decomposition of width at most one.*

Proposition 2.9 is the mathematically honest containment theorem behind the intended “tree structure is the treewidth-one special case” slogan. It distinguishes this parent-tree notion from the weaker monotone-order predicate currently named **TreeStructured**.

Proposition 2.10¹⁰ (Current Positive Interfaces Factor Through Role Classification). *The current positive theorem surface factors through the role classifier of Theorem 2.1. Concretely, the family-indexed interfaces factor*

1. *subcase-facing tractability,*
2. *degenerate-case witnesses, and*
3. *complexity-class summaries*

all factor through the map

$$\text{family} \longmapsto \text{role} \in \{\text{core}, \text{lifted}, \text{degenerate}\}.$$

Proposition 2.10 is the formal reason the current landscape feels smaller than its file count. The exported positive API is already controlled by the role partition rather than by fifteen unrelated theorem schemas.

3 Lifts and Degeneracies

The extra named families beyond the six core subcases split naturally into two groups.

Regime Lifts

Four stochastic/sequential positive results do not introduce new primitive mechanisms. They reduce directly to existing core subcases:

1. product distributions reduce to separable utility;
2. bounded support reduces to bounded actions;
3. bounded horizons reduce to bounded treewidth;
4. full observability reduces to tree structure.

These are mathematically useful because they transfer the static tractable theory into richer regimes. But they do not enlarge the primitive basis. They are best understood as *lifts*, not new frontier points.

Proposition 3.1¹¹ (Lifted Cases Introduce No New Primitive). *Each stochastic/sequential tractable case considered here is classified into one of the six core structural subcases. In particular, these positive results do not enlarge the primitive basis from Corollary 2.3.*

⁸ : FR115 ⁹ : FR100, FR110–111 ¹⁰ : FR4 ¹¹ : FR5

Degenerate Cases

The remaining positive cases considered here are degenerate in a precise sense.

Single-action systems, strict global dominance, constant optimal-set systems, and multiplicative-separable constant-sign models all collapse the optimizer so strongly that the relevant-coordinate question disappears: the optimal-action set is constant across states, so the empty coordinate set is sufficient. These are not new structural tractability mechanisms for the relevance problem; they are constant-optimizer collapses.

Bounded state space behaves differently. It does not force a constant optimizer. Instead, it makes exact certification feasible by brute-force enumeration over a finite universe. This is polynomial only because the instance family already carries a hard external bound on its search space.

Proposition 3.2¹² (Degenerate Positive Cases). *Every non-core tractable case considered here that is not a regime lift belongs to one of two degenerate buckets:*

1. *constant-optimizer collapse, or*
2. *finite explicit enumeration.*

Taken together with Proposition 3.1, Proposition 3.2 sharpens Theorem 2.1: every positive result considered here is exactly one of three things, a primitive core case, a lift, or a degenerate collapse.

4 Toward the Frontier Theorem

Sections 2 and 3 motivate a frontier question. Do the tractable families already suggest a finite optimizer-compatible criterion? In the strongest form, could exact relevance certification admit a representation-sensitive dichotomy theorem analogous to the classical dichotomy programs for satisfiability and constraint satisfaction?

This section shows that the most direct version of that program fails. The realizability barrier shows that quotient shape is too expressive, while the admissibility-layer collapse theorem later in the section shows that even structurally restricted finite classifiers fail on a theorem-forced surface. What remains is a sharper statement of what any future frontier theorem must avoid.

The obstacles are not all of the same kind. First, realizability is not the bottleneck: Theorem 5.1 shows that arbitrary labeling kernels already arise as optimizer quotients, so no frontier theorem can be driven by quotient realizability alone. Second, even on the positive side one must separate primitive mechanisms from lifts and degeneracies; otherwise a finite inventory confuses genuine sources of tractability with derived or collapsed cases. Third, the long-term target remains an algebraic description analogous to the CSP dichotomy program, but the negative results below show that no straightforward optimizer-compatible classifier can play that role. Section 6 isolates the necessary conditions that survive these three obstacles.

A Stabilized Binary Pairwise Subregime

One piece of the frontier program already admits a clean canonical theorem. On binary coordinate domains, define the mixed difference along coordinates i, j for action a by evaluating the utility on the four assignments $(0, 0), (1, 1), (0, 1), (1, 0)$ while holding all other coordinates fixed at 0. For pairwise utilities, this mixed difference is the canonical witness of genuine pair interaction.

Proposition 4.1¹³ (Binary Pairwise Symmetry Dichotomy). *Let the coordinate alphabet be binary. If a utility admits a pairwise decomposition and is invariant under coordinate permutations, then either:*

¹² : FR1, FR6 ¹³ : FR118-120

1. *it admits a unary-coordinate decomposition, so every pairwise term collapses into single-coordinate contributions, or*
2. *every distinct coordinate pair carries a genuine pair interaction, witnessed by a nonzero mixed difference for some action.*

The point is structural rather than cosmetic. Within the binary pairwise-symmetric regime, there is no intermediate sparse interaction pattern hiding between unary collapse and the complete-pair case. Once a single nonzero pair witness exists, symmetry propagates it to every coordinate pair. This removes one plausible source of additional primitive mechanisms from the frontier search.

The optimizer-relevant version needs one extra layer. Raw pair interaction is still too coarse, because action-independent pairwise terms can be structurally dense without changing the optimizer. The correct invariant is the mixed difference of action gaps $U(a, \cdot) - U(b, \cdot)$.

Proposition 4.2¹⁴ (Decision-Relevant Binary Pairwise Dichotomy). *In the same binary pairwise-symmetric regime, either all action-dependent pair effects collapse into a unary-coordinate reduction after removing an action-independent base state term, or every distinct coordinate pair is decision-relevantly interacting for some action gap.*

This is the relevant replacement for a coarse hardness heuristic. The obstruction theorem shows that even complete genuine pair interaction and failure of coordinate symmetry do not force hardness by themselves: one can still have a constant optimizer. Any eventual frontier theorem must therefore quotient out optimizer-degenerate phenomena, not just declared or utility-level interaction density.¹⁵

The obstruction is stronger than that first formulation suggests. Action-specific utility offsets leave the decision-relevant interaction graph unchanged, yet they can turn a family with nonconstant optimizer behavior into one with a constant optimizer while preserving dense decision-relevant structure. Any true hardness theorem in this regime must therefore normalize away action offsets explicitly, not merely require dense decision-relevant interaction.¹⁶

This normalization can be stated cleanly: utilities are identified up to action-dependent, state-independent offsets, and the offset-normalized decision-relevant graph is well-defined on those equivalence classes. But the resulting dichotomy attempt still fails. The offset-normalized obstruction theorem shows that even after passing to offset classes, one can retain complete decision-relevant interaction and unbounded treewidth while exact certification remains trivial.¹⁷

The next refinement is to restrict the interaction graph to optimizer-supported actions. This removes two earlier collapse mechanisms: the offset-collapse family and a ghost-action family whose all-action decision-relevant graph is complete even though a single coordinate remains sufficient. But even that support-filtered graph is not enough. The optimizer-supported obstruction theorem exhibits a margin-masking family with complete optimizer-supported decision-relevant interaction and unbounded treewidth, yet the optimizer still depends only on coordinate 0. The third collapse mechanism is therefore not ghost support or offset collapse, but large unary margins that mask dense supported pairwise interactions.¹⁸

One might hope that a strict unary-to-pair margin bound rescues the dichotomy. Define *margin-bounded* pairwise utilities by requiring every unary term to be at most twice the largest binary mixed-difference magnitude. This still fails. The dominant-pair family is margin-bounded because its unary terms vanish, while its optimizer-supported decision-relevant graph remains complete. Yet the obstruction theorem shows that exact certification is still controlled by the two-coordinate set $\{0, 1\}$. The fourth collapse mechanism is therefore pair-weight concentration rather than unary masking: one supported pair dominates the optimizer while the remaining dense interactions are too weak to matter.¹⁹

¹⁴ : FR121-124 ¹⁵ : FR124 ¹⁶ : FR125-127 ¹⁷ : FR128-129 ¹⁸ : FR130-136 ¹⁹ : FR137-140

Admissible Closure-Orbit No-Go Program

The obstruction families above lead to a uniform negative statement. The decisive fact is that they admit explicit disagreements inside a single closure orbit.

Proposition 4.3²⁰ (Orbit-Gap Template). *Let Q be a slice predicate. Suppose there exist slices U, V such that U and V lie in the same closure orbit, $Q(U)$ holds, and $Q(V)$ fails. Then no closure-law-invariant predicate P can satisfy*

$$P(W) \iff Q(W) \quad \text{for all slices } W.$$

Proof idea. Closure-law invariance gives $P(U) \iff P(V)$. Exact agreement with Q would then imply $Q(U) \iff Q(V)$, contradicting the assumed orbit gap. ■

The proof is constructive: for each obstruction family, one exhibits a same-orbit disagreement witness. In all four cases the witness is a positive-affine step whose α -component is an action-independent state term supported on a single coordinate pair. This is the central technical mechanism. Pure relabeling cannot change the relevant obstruction statistics, and a global scale factor preserves all pairwise magnitudes up to common rescaling. The statewise additive term is what allows one to move mass onto a chosen pair while remaining inside the theorem-forced closure laws. The fact that the same mechanism works for four structurally different obstruction families is strong evidence that the collapse theorem captures a genuine phenomenon rather than a collection of unrelated counterexamples.

There is also a transport interpretation of these witnesses. Once transport is computed on optimizer-quotient classes, a singleton quotient admits zero-cost diagonal transport, whereas genuine branching forces positive off-diagonal transport when distinct classes carry mass. The pair-targeted affine witnesses exploit exactly this sensitivity. They do not introduce arbitrary perturbations unrelated to the decision geometry; rather, they move mass across a selected branch of the quotient while staying inside the same closure orbit. In this sense the affine witnesses track the intrinsic transport geometry of the quotient instead of functioning as purely syntactic tricks.²¹

Theorem 4.4²² (Dominant-Pair Admissible No-Go). *No admissible normalization predicate can decide the dominant-pair obstruction family exactly.*

Proof sketch. The target predicate is unique dominant-pair status: one pair/action realizes the maximal interaction magnitude and that maximizer is unique. Start from a slice whose unique dominant pair is $\{0, 1\}$. Add an action-independent affine state term supported on a different pair, say $\{1, 2\}$. This produces a slice in the same closure orbit but with a different unique dominant pair. Proposition 4.3 then rules out every closure-law-invariant classifier, hence every admissible one. ■

Theorem 4.4 is the canonical instance. The remaining three families use the same orbit-gap scheme, but each isolates a different structural obstruction.

Theorem 4.5²³ (Margin-Masking Admissible No-Go). *No admissible normalization predicate can decide margin-boundedness exactly.*

Proof sketch. For margin masking, the target predicate is margin-boundedness itself. This family is structurally different from the dominant-pair family because it is governed by a threshold comparing unary mass against the largest pair interaction. The affine witness leaves unary terms unchanged but raises the largest pair interaction past the relevant threshold, so the translated slice becomes

²⁰ : FR185 ²¹ : FR183–184 ²² : FR186 ²³ : FR187

margin-bounded while the base slice is not. Proposition 4.3 therefore rules out every closure-law-invariant classifier, hence every admissible one. ■

Theorem 4.6²⁴ (Ghost-Action Admissible No-Go). *No admissible normalization predicate can decide the ghost-action concentration signature exactly.*

Proof sketch. For ghost actions, the target predicate says that there is an action whose unary contribution on the first coordinate is -1 on both binary values and whose anchor-pair interaction has unit mixed-difference magnitude. This family is local in appearance but unstable under the same pair-targeted affine move. A pair-supported affine term preserves the closure orbit while destroying the signature, so Proposition 4.3 applies. ■

Theorem 4.7²⁵ (Offset Admissible No-Go). *No admissible normalization predicate can decide the additive/statewise offset signature exactly.*

Proof sketch. For additive/statewise offset concentration, the target predicate requires two actions with distinct anchor-pair interaction magnitudes, namely 1 and 0. This family isolates the action-specific mismatch that survives after earlier offset normalizations. A pair-supported affine term changes the anchor-pair statistics while remaining in the same closure orbit. Again, Proposition 4.3 yields the contradiction. ■

Theorem 4.8²⁶ (Admissible Collapse Across the Four Obstruction Families). *Let P be a normalization predicate satisfying the admissibility axioms. Then P is not a tractability characterization.*

Proof idea. If P were an admissible tractability characterization, it would have to decide each of the four obstruction predicates above. Theorem 4.4, Theorem 4.5, Theorem 4.6, and Theorem 4.7 show that this is impossible. Hence no admissible normalization predicate yields the desired frontier theorem. ■

5 The Realizability Barrier

One possible route to a finite tractability taxonomy would be to show that optimizer-induced quotients realize only a narrow subclass of equivalence relations, so that the frontier would emerge from realizability alone. In the unconstrained setting, this route fails.

Theorem 5.1²⁷ (Every Labeling Kernel Is Optimizer-Realizable). *Let $\phi : S \rightarrow T$ be any function. There exists a decision problem with action space T and state space S such that for every state $s \in S$,*

$$\text{Opt}(s) = \{\phi(s)\}.$$

Consequently, for all states $s, s' \in S$,

$$\text{Opt}(s) = \text{Opt}(s') \iff \phi(s) = \phi(s').$$

Equivalently, the decision quotient of the constructed problem is exactly the kernel partition of ϕ .

Proof. Take the action set to be T itself and define

$$U(a, s) = \begin{cases} 1 & \text{if } a = \phi(s), \\ 0 & \text{otherwise.} \end{cases}$$

²⁴ : FR188 ²⁵ : FR189 ²⁶ : FR190 ²⁷ : FR7-8

Then the unique maximizer at state s is the designated action $\phi(s)$, so the optimizer set is exactly $\{\phi(s)\}$. Two states therefore have the same optimizer set if and only if they receive the same label under ϕ . ■

Theorem 5.1 is the key negative fact for the frontier program. It shows that optimizer realizability by itself is extremely permissive. If arbitrary label kernels are already optimizer-induced, then no finite taxonomy can come merely from asking which abstract quotients are realizable.

Proposition 5.2²⁸ (Every Equivalence Relation Is Optimizer-Realizable). *For every equivalence relation on the state space, there exists a decision problem whose decision quotient relation is exactly that equivalence relation. Moreover, the quotient object of the realizing problem is canonically equivalent to the corresponding setoid quotient.*

Proof. Let $\approx$ be an equivalence relation on S , and let $\pi : S \rightarrow S/\approx$ be the quotient map. Apply Theorem 5.1 to the labeling map π . The resulting decision problem satisfies

$$\text{Opt}(s) = \text{Opt}(s') \iff \pi(s) = \pi(s'),$$

and the right-hand side is exactly the original equivalence relation. Since the realizing quotient identifies states precisely by equality of quotient labels, its quotient object is canonically equivalent to the setoid quotient itself. ■

Proposition 5.2 is the maximal realizability statement available in the present framework. The labeling-kernel theorem is a special case. Thus the realizability barrier is not merely broad; at the level of abstract quotient relations, it is maximal.

Proposition 5.3²⁹ (The Realized Quotient Is Exactly the Label Range). *For every labeling map $\phi : S \rightarrow T$, the decision quotient of the realizing problem is canonically equivalent to the range of ϕ . In finite settings, the quotient cardinality is therefore exactly $|\text{range}(\phi)|$.*

Proof. By Theorem 5.1, two states lie in the same decision class if and only if they receive the same label under ϕ . Therefore the quotient class of a state depends only on the value $\phi(s)$, and sending the class of s to $\phi(s)$ defines a well-defined map from the decision quotient to $\text{range}(\phi)$. This map is surjective by definition of the range and injective because equality in the range means equality of labels, hence equality of quotient classes. ■

Proposition 5.3 strengthens Theorem 5.1. The realizing construction does not merely realize the induced equivalence relation abstractly; it realizes the quotient object itself with exactly the expected image size.

Corollary 5.4³⁰ (Realizability Alone Cannot Be the Frontier). *Any future optimizer-compatible tractability metatheorem must use more than quotient realizability alone. Additional restrictions have to enter through representation, coordinate structure, utility structure, closure properties of the family under study, or comparable algebraic invariants.*

This shifts the burden of the metatheorem. The hard question is not which quotients can arise in principle; almost all kernels can. The hard question is which *natural structural families of decision problems* force tractability or preserve hardness once coordinate structure and utility representation are taken seriously.

²⁸ : FR45–47 ²⁹ : FR39–40 ³⁰ : FR7–8

6 Necessary Conditions on Any Frontier Statement

The realizability barrier leaves a more precise frontier question. If quotient shape alone is too expressive, what should replace it as the organizing unit of the metatheorem?

At minimum, the tractability frontier should not classify arbitrary sets of decision problems. It should range over representation-level structural classes and it should ignore benign changes of presentation.

The right eventual condition list will almost certainly be stronger than this minimal principle. For example, a mature frontier theorem may also require closure under adding explicitly irrelevant coordinates, adjoining dummy actions, or other representation-preserving refinements. But action/state relabel invariance is the irreducible starting point.

Proposition 6.1³¹ (Exact Certification Depends Only on the Decision Quotient Relation). *If two decision problems on the same state space induce the same decision-equivalence relation, then they induce the same exact-certification structure. In particular, they have the same sufficient coordinate sets, the same minimal sufficient coordinate sets, and the same relevant and irrelevant coordinates. Moreover, their quotient objects are canonically equivalent; in finite settings, those quotient objects therefore have the same cardinality.*

Proposition 6.1 is the canonical statement behind the frontier program. Exact certification is determined by the decision quotient relation itself. Equality of optimizer maps is only a sufficient route to this conclusion, because equal optimizer maps induce the same decision-equivalence relation.

Corollary 6.2³² (Sufficiency Is Relation Refinement). *A coordinate set I is sufficient if and only if the coordinate-agreement relation induced by I refines the decision quotient relation.*

Corollary 6.2 is the cleanest relational form of exact certification. It removes utility language entirely: sufficiency asks whether one state relation is contained in another.

Corollary 6.3³³ (Relevance Is Failure of Erased Sufficiency). *A coordinate i is irrelevant if and only if deleting it leaves a sufficient coordinate set, namely $\text{univ} \setminus \{i\}$. Equivalently, i is relevant if and only if that erased set is not sufficient.*

Corollary 6.3 makes the role of individual coordinates canonical: relevance is exactly the obstruction to retaining sufficiency after coordinate erasure.

Corollary 6.4³⁴ (Certification Statistics Factor Through the Quotient Relation). *Any statistic that is a function of the sufficient-set family and relevant-coordinate set is invariant under equality of the decision quotient relation. In particular, the number of sufficient coordinate sets, the number of minimal sufficient coordinate sets, and the number of relevant coordinates are quotient invariants.*

Corollary 6.4 records the next logical consequence of Proposition 6.1. Once exact certification is identified with quotient data, every certification statistic built from sufficient sets or relevant coordinates factors automatically through that quotient data as well.

Proposition 6.5³⁵ (Zero-Distortion Summaries Refine the Optimizer Quotient). *Let $\sigma : S \rightarrow C$ be a summary map. Suppose that*

$$\sigma(s) = \sigma(s') \implies \text{Opt}(s) = \text{Opt}(s') \quad \text{for all } s, s' \in S.$$

Then each summary fiber is contained in a single optimizer-quotient class. Equivalently, σ factors through the optimizer quotient relation, and distinct optimizer classes require distinct summary symbols.

³¹ : FR30–35, FR37, FR51 ³² : FR48 ³³ : FR49–50 ³⁴ : FR41–44 ³⁵ : FR176–177

Proof. If $\sigma(s) = \sigma(s')$, the hypothesis gives $\text{Opt}(s) = \text{Opt}(s')$. So every σ -fiber is contained in one decision-equivalence class. This is exactly the statement that σ refines the optimizer quotient. Conversely, if two distinct optimizer classes shared a summary symbol, choosing representatives from those classes would violate the zero-distortion premise. ■

Proposition 6.5 explains why exact relevance is not merely one point on an approximation spectrum. Once distortion is required to be zero, lossless decision-preserving abstraction is already constrained by the optimizer quotient itself. Approximate summaries may still be useful for other goals, but they no longer certify exact preservation of the decision boundary.

Corollary 6.6³⁶ (Constant-Optimizer Collapse Forces Trivial Certification). *If the optimizer set is constant across all states, then the empty coordinate set is sufficient and every coordinate is irrelevant.*

Corollary 6.6 isolates the first genuinely degenerate mechanism in the tractable basis. The certification problem disappears because there is no state dependence left to certify.

Proposition 6.7³⁷ (Explicit Enumeration Is Parameter-Dependent, Not Structural). *For every fixed exponent k , there exists a Boolean-cube family with state space size $2^n > n^k$. Thus explicit-state enumeration can outrun any fixed monomial bound in the ambient dimension parameter.*

Proposition 6.7 is weaker than a full polynomial lower-bound metatheorem, but it already captures the structural point needed here: tractability by brute-force state enumeration depends on a separate size parameter and does not arise from an optimizer-compatible mechanism like symmetry, low rank, or tree structure.

Proposition 6.8³⁸ (Positive Affine Utility Reparameterizations Are Invisible). *Statewise positive affine transformations of utility leave exact certification unchanged. In particular, replacing*

$$U(a, s) \quad \text{by} \quad \alpha(s) + \beta(s)U(a, s)$$

with $\beta(s) > 0$ for every state s does not change sufficient coordinate sets or relevance judgments.

Proposition 6.8 is another direct consequence of Proposition 6.1. It records a standard utility-theoretic invariance in the exact-certification language: only the induced decision quotient matters, not the particular positive affine scale used to encode utility.

Proposition 6.9³⁹ (Duplicate Actions and Duplicate States Preserve Certification). *Duplicating a single action without changing its utility profile preserves the decision quotient relation and hence preserves sufficiency and relevance. At the optimizer-set level, the only possible change is the addition of the duplicate action itself: original optimal actions remain optimal, and the duplicate is optimal exactly when the original action was. Duplicating a single state without changing its utility profile preserves the decision quotient up to the obvious quotient equivalence and also preserves sufficiency and relevance.*

Proposition 6.9 gives two more theorem-forced closure laws. These are not modeling conventions; they are consequences of the fact that exact certification factors through quotient data rather than through accidental multiplicity in the presentation.

Proposition 6.10⁴⁰ (Relabeling Invariance Is Forced by Exact Certification). *Action relabeling preserves optimizer equivalence and exact sufficiency. State relabeling does so as well once the coordinate structure is transported along the state bijection. Consequently, any structural tractability frontier for exact relevance certification must be closed under these relabelings.*

³⁶ : FR52–53 ³⁷ : FR66–67 ³⁸ : FR13–16 ³⁹ : FR54–56, FR68–73 ⁴⁰ : FR17–20

Proposition 6.10 is now best read as a corollary of Proposition 6.1. Bijective relabelings do not change the optimizer map up to the obvious identification, so they cannot change exact certification.

The second necessary ingredient is a genuine expressivity restriction. Call a class *kernel-universal* if, for every labeling map $\phi : S \rightarrow T$, it contains some decision problem whose optimizer quotient realizes exactly the kernel partition of ϕ . By Theorem 5.1, kernel universality is an extremely strong property.

The canonical finite invariant here is the size of the optimizer quotient, equivalently the number of distinct optimal-action sets. The quotient is identified with the range of the optimizer map, so quotient size is not auxiliary bookkeeping; it is intrinsic.

Proposition 6.11⁴¹ (Quotient Size Is Unbounded Under Realizability). *For every $m \in \mathbb{N}$, there exists a finite decision problem whose optimizer quotient has size exactly m .*

Proof. Take $m = k + 1$ and realize the identity labeling on $\text{Fin}(k + 1)$. Proposition 5.3 identifies the quotient object with the range of that identity labeling, which has cardinality $k + 1$. Equivalently, Proposition 5.2 realizes the discrete equivalence relation on $\text{Fin}(k + 1)$. ■

The point is simple: a meaningful frontier statement cannot range over classes that are simultaneously relabel-invariant and kernel-universal, because quotient shape alone then becomes too expressive to isolate the boundary.

Proposition 6.12⁴² (Invariance Alone Is Too Weak). *The universal class of all decision problems satisfies action relabeling invariance and state relabeling invariance, but it is kernel-universal. Consequently, relabeling invariance by itself is far too weak to isolate a tractability frontier.*

Proposition 6.12 separates two kinds of conditions that can otherwise be conflated.

1. **Benign invariance axioms** say the family should ignore arbitrary naming choices.
2. **Expressivity-limiting axioms** say the family should not realize arbitrary quotient structure wholesale.

Any genuine optimizer-compatible frontier theorem will need both. Without decision-quotient invariance and its immediate corollaries such as relabel invariance and positive affine invariance, the statement is not structural. Without an expressivity restriction excluding kernel universality or something comparably strong, the realizability theorem and Proposition 6.11 make the class too large for quotient shape alone to carry the boundary.

Proposition 6.13⁴³ (Independent Summary Frameworks Converge on the Same Structural Core). *Write $r = \text{srnk}(\mathcal{D})$ for the number of relevant coordinates of a Boolean decision problem, and let m be the number of optimizer-quotient classes. Then:*

1. $m \leq 2^r$, so the counting entropy $\log_2 m$ is at most r ;
2. the diagonal 0/1 relevance-support matrix has rank exactly r ;
3. every zero-distortion decision-preserving summary requires at least m distinct summary symbols.

Hence quotient entropy, Fisher-style support counting, and zero-distortion summary size are all controlled by the same support/quotient core.

⁴¹ : FR24, FR38 ⁴² : FR21–23 ⁴³ : FR178–182

Proof. If two states agree on all relevant coordinates, then they differ only on irrelevant ones. Changing those irrelevant coordinates one at a time cannot change the optimizer, so the relevant coordinates already determine the optimizer class. Therefore there are at most 2^r optimizer classes, one for each Boolean assignment on the relevant support, and $\log_2 m \leq r$ follows.

For the second clause, the diagonal relevance-support matrix has a 1 exactly on relevant coordinates and a 0 elsewhere. Its rank is therefore the number of relevant coordinates, namely r .

The third clause is exactly Proposition 6.5: a zero-distortion summary must assign distinct symbols to distinct optimizer classes. ■

The point of Proposition 6.13 is not that entropy, support counting, and lossless summaries become identical notions. The point is that they already converge on the same structural content before any admissibility axiom is imposed. This is why closure-law invariance is the decisive semantic clause in the admissibility class below: once several independent frameworks ignore surface presentation in favor of the support/quotient core, a structural tractability classifier should do the same. The remaining admissibility clauses serve as algorithmic and semantic guardrails.

From Necessary Conditions to an Admissibility Class

The closure properties above are theorem-forced by exact-certification semantics, but they are not enough on their own to support a meaningful impossibility theorem. If one allows unrestricted predicates, a semantic predicate can simply encode the target complexity class and trivialize the statement.

Proposition 6.14⁴⁴ (Unrestricted Predicates Trivialize Exact Characterization). *Without admissibility restrictions, there exist normalization predicates whose membership condition directly encodes the intended polynomial-time side of the landscape, so unrestricted collapse-impossibility claims fail.*

Proposition 6.14 is not a by-product of the proof method. It is the expected behavior of unrestricted semantic predicates. Therefore any meaningful frontier theorem must be parameterized by an admissibility class that simultaneously (i) respects theorem-forced invariances and (ii) blocks direct semantic encoding.

Definition 6.15 (Admissible Normalization Predicate). A normalization predicate is *admissible* when it satisfies all of the following:

1. polynomial-time checkability at the slice level;
2. invariance under the closure laws forced by exact certification (relabelings, positive affine reparameterizations, duplication operations, and irrelevant-coordinate extension);
3. structural extractability of the associated dependency graph;
4. bounded-pattern definability.

More explicitly, let $|U|$ denote the encoding size of a binary pairwise slice U , and let $X(U)$ be its canonical finite pairwise syntax. Polynomial-time checkability means that there exist constants $c, k \in \mathbb{N}$ and a decision procedure A such that

$$A(U) = 1 \iff Q(U), \quad \text{time}_A(U) \leq c(|U| + 1)^k + c.$$

Structural extractability means that whenever $Q(U)$ holds, the associated coordinate graph is obtained from syntax alone: there exists a uniform extractor

$$E : X \mapsto G_E(X)$$

⁴⁴ : FR145

on finite pairwise syntactic presentations such that the graph attached to U is exactly $G_E(X(U))$.

Bounded-pattern definability is likewise finite and explicit. There must exist uniform bounds on radius, neighborhood size, action alphabet size, and coefficient magnitude, together with finite lists of permitted and forbidden rooted local patterns, such that membership in Q is determined by the presence of one witness pattern or the absence of all forbidden patterns in the rooted interaction neighborhoods of $X(U)$. No unbounded global computation is hidden inside the definition.

Each clause in Definition 6.15 has an independent methodological role.

Polynomial-time checkability is the minimal algorithmic requirement. A tractability characterization that cannot itself be recognized efficiently has little explanatory value: it would only relocate the computational difficulty from exact certification to the membership test for the classifier.

Closure-law invariance is the decisive semantic requirement. It is not stipulated for convenience. It is forced by the Block 6 invariance theorems, because exact certification itself is unchanged by relabelings, positive affine utility reparameterizations, duplication operations, and irrelevant-coordinate extensions. The orbit-witness arguments of Section 4 use precisely this clause and no stronger semantic assumption.

Structural extractability distinguishes structural characterizations from purely semantic ones. Without such a condition, one can define predicates directly on optimizer behavior or on the solved certification instance itself, bypassing the stated aim of classifying tractability by representation-level structure.

Bounded-pattern definability is the locality guardrail. Its role is to exclude predicates whose membership test secretly performs an unbounded global computation under a structural name. In particular, it prevents one from reintroducing oracle power through an arbitrarily large finite schema.

Definition 6.15 is therefore not a list assembled solely to support the impossibility theorem. Closure-law invariance supplies the semantic content of the no-go theorem; the remaining clauses exclude vacuous or non-structural counterexamples and keep the classification problem within the intended algorithmic setting.

7 Related Work

Rough Sets, Feature Selection, and Exact Relevance

At the static level, the closest classical neighbor is rough-set reduct theory [19, 30]. That literature studies which attributes can be deleted while preserving decision distinctions. Exact static sufficiency can be viewed as a reduct condition for the induced decision table. The present paper is different in scope: it is not a new reduct algorithm, but a structural study of which tractable mechanisms admit a finite characterization.

A separate neighboring literature studies feature selection, variable importance, and model explanation in machine learning and AI, including wrapper and filter methods [15], general surveys [13], and attribution methods based on Shapley-style decompositions [16]. These works are relevant by motivation rather than by technique. They typically optimize predictive performance, explanatory salience, or approximation quality under data-dependent criteria. Exact relevance certification instead asks for a semantic preservation property: which coordinates are necessary to preserve the optimizer relation exactly.

A further nearby tradition is decision-theoretic informativeness and value of information, beginning with Blackwell’s comparison of experiments and subsequent decision-theoretic treatments [4, 23, 14]. The connection is conceptual: both settings ask when one representation retains

all decision-relevant distinctions of another. The difference is that the present paper studies the combinatorial complexity of certifying exact preservation under coordinate deletion.

Information-Theoretic and Transport Viewpoints

The zero-distortion discussion in this paper belongs to the classical information-theoretic tradition of exact distinguishability and lossless coding boundaries, going back to Shannon and standard modern treatments of source coding and rate-distortion theory [25, 26, 27, 6]. Fisher-style support counting supplies another classical comparison lens on relevant support [8]. Our use of these viewpoints is limited and structural. The paper does not formulate a general coding or statistical estimation problem. It uses zero distortion, entropy, and support counting only to show that several independent summary frameworks are already constrained by the same optimizer-quotient core. The Wasserstein language in Section 4 is used in the same spirit: as a qualitative geometric description of quotient branching, not as a full transport-theoretic model.

Backdoors, Tractable Islands, and Dichotomy Programs

The nearest complexity-theoretic analogue is backdoor tractability for constraint satisfaction [31, 3, 9]. There, a small structural set exposes membership in a tractable class even when the ambient problem is hard. The comparison is direct: exact relevance certification also asks which coordinates matter, and polynomial-time behavior emerges when the source of hardness is restricted by structure. Bessiere et al. [3] make especially clear that exploiting a known tractable structure and discovering the responsible structure are different algorithmic tasks; later parameterized work sharpened that distinction. This is closely aligned with the present paper’s separation between positive tractable mechanisms and the harder meta-level problem of recognizing a correct frontier classifier.

The broader analogy is the dichotomy tradition for satisfiability and constraint satisfaction, from Schaefer’s Boolean dichotomy theorem [24] to the finite-domain CSP dichotomy theorems of Bulatov and Zhuk [5, 34]. The expectation that a clean tractability boundary should exist was articulated in the Feder–Vardi program [7], and later algebraic work of Barto and Kozik clarified why finite structural boundaries are plausible in that setting [2]. Against that background, the present paper does not argue against dichotomy theorems in general. It shows instead that, for exact relevance certification, quotient realizability and closure-law invariance leave too much room for an admissible optimizer-compatible classifier of the most direct kind to survive. Any successful frontier theorem in this domain must therefore use stronger representation-sensitive structure.

The quotient viewpoint also touches the literature on exact abstraction and aggregation in stochastic control, Markov decision processes, and reinforcement learning [18, 11, 20]. Those works study when states may be aggregated while preserving value or policy structure. Our setting is narrower: we ask for exact preservation of optimizer classes under coordinate-hiding maps rather than arbitrary state abstractions. Still, the common theme is that semantic invariants of decision behavior should survive presentation-level simplification.

Meta-Impossibility Traditions

The negative theorem in this paper belongs to a broader family of impossibility results. At the foundational level, Gödel’s incompleteness theorems show that sufficiently expressive formal systems cannot internalize all truths about their own proof theory [12]. In learning and optimization, no-free-lunch results show that universal guarantees disappear on overly broad problem domains [32, 33]. In social choice, Arrow’s theorem and the Gibbard–Satterthwaite theorems show that natural axiomatic requirements become jointly unsatisfiable on rich enough domains [1, 10, 22]. The present

theorem has the same broad form: one fixes a principled invariance package and then proves that no exact classifier can satisfy it on the intended domain.

The closest analogy is nevertheless Rice-style impossibility. Rice’s theorem [21] shows that nontrivial extensional properties of sufficiently expressive semantic objects are undecidable. Refinements such as the Myhill–Shepherdson theorem and the Rice–Shapiro theorem show that this is not a single isolated fact but part of a broader pattern for structured semantic domains [17, 28]. The analogue in the present paper is that dominant-pair concentration, margin-boundedness, ghost-action concentration, and additive/statewise offset concentration are nontrivial slice predicates that are invariant under the theorem-forced closure laws.

The Arrow parallel is complementary. The admissibility package plays the role of the axiom system: closure-law invariance formalizes semantic irrelevance, while polynomial-time checkability, structural extractability, and bounded-pattern definability exclude semantic encodings that would make the statement vacuous. Once those requirements are imposed, the orbit-gap witnesses show that exact correctness is impossible. The result is therefore a structural impossibility theorem for a natural class of representation-sensitive classifiers.

8 Conclusion

This paper establishes both sides of the frontier question. On the positive side, the tractable landscape studied here factors into primitive mechanisms, lifts, and degeneracies. On the negative side, quotient realizability is maximal, so quotient shape cannot classify tractability by itself, and an admissibility-layer collapse theorem rules out direct finite classifiers even after imposing theorem-forced closure laws together with computational and structural restrictions.

Technically, the decisive mechanism is closure-orbit disagreement via pair-targeted action-independent affine transport. The same template now applies to four obstruction families (dominant pair, margin masking, ghost action, and additive/statewise offset), yielding an admissible-collapse theorem.

The open problem is therefore sharper than before. The question is no longer whether simple finite lists or unconstrained quotient predicates can characterize the boundary; they cannot. The research program is redirected toward a narrower target: identify a stronger representation-sensitive principle, outside the present admissible surface but still mathematically natural, that could support a genuine optimizer-compatible dichotomy. In that sense, the meta-impossibility theorem is not only a negative result. It determines which kinds of frontier theorem are no longer worth pursuing.

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